

BINGXI (PAUL) JIANG

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EDUCATION

University of California, Irvine (UCI) | Irvine, CA

Bachelor's Degree, Double Major: Computer Science; Business Information Management | GPA: 3.725/4.0 | Expected June 2027

Relevant Coursework: Artificial Intelligence, Machine Learning, Database Management, Human Computer Interaction, Information Retrieval

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, JavaScript, Java, Bash

ML/DL Frameworks: PyTorch, TensorFlow, Scikit-Learn, Hugging Face, XGBoost

MLOps & Tools: MLflow, TensorBoard, Weights & Biases, LLaMA-Factory, ONNX

Backend & APIs: Flask, FastAPI, Node.js, Express, REST APIs

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis

Cloud & DevOps: AWS (EC2, S3, SageMaker), Docker, Kubernetes, Git, Linux

Data Processing: Pandas, NumPy, PySpark, Airflow

Monitoring: Prometheus, Grafana

Other: LoRA, Prompt Engineering, Tool Orchestration, A/B Testing, Canary Deployment

PROFESSIONAL EXPERIENCE

Fresh Road Inc | San Jose, CA

Software Engineer Intern | September 2025 – December 2025

- Architected production backend agent framework with **Node.js** integrating **OpenAI** and **Gemini** implementing tool schemas and templates.
- Optimized cost by 32% and p95 latency by 18% implementing caching, compression and selective routing with model tiers.
- Engineered B2B AI-agent callback pipeline adding ownership attribution and transcript tracking improving operational efficiency by 50%.
- Developed evaluation tooling including regression suites and golden conversations enabling A/B testing and monitoring.
- Deployed services with **Docker** containers on **AWS EC2** implementing load balancing and auto-scaling handling 10K+ requests daily.

Ping An Technology | Guangdong, China

Computer Vision Algorithm Engineer | June 2025 – September 2025

- Trained **Qwen2.5-VL** for vehicle-damage detection on 2.27M samples using **LoRA** with **LLaMA-Factory** achieving 69% precision.
- Designed automated damage-reporting system with prompt orchestration and tool integration reducing manual review by 20%.
- Accelerated inference performance with batching and mixed precision reducing p95 latency by 38% and increasing throughput by 50%.
- Deployed model with canary rollout and **MLflow** registry implementing monitoring and rollback ensuring safe production deployment.
- Enhanced low-quality images using **CLIP**-semantic residual guidance with adaptive equalization improving recall by 7 points.

Relational Cognition Lab | Irvine, CA

Research Assistant | January 2026 – Present

- Conducted representation probing on frontier LLMs including **Gemini**, **GPT-4** and **Llama-3** evaluating internal consistency across layers.
- Developed noise-estimation framework interspersing Wikipedia text in prompts establishing statistical baselines for vector projections.
- Architected experiment infrastructure for dataset curation, batched inference and metrics dashboards hardening pipelines against drift.
- Implemented evaluation pipeline with **PyTorch** processing 100K+ prompt-response pairs tracking similarity and layer activations.
- Deployed analysis infrastructure with **Docker** and **PostgreSQL** storing experiment results and metadata enabling reproducible workflows.

DeepEM Lab | Irvine, CA

Research Intern | January 2025 – June 2025

- Constructed **Scikit-Learn** baselines with PCA and SVM predicting lithium-ion performance from material ratios delivering pipelines.
- Conducted cross-validation evaluation with hyperparameter tuning achieving 87% accuracy on test set providing interpretable boundaries.
- Engineered data preprocessing pipeline with **Pandas** handling missing values, outliers and feature scaling improving performance by 15%.
- Implemented automated experiment tracking with **MLflow** logging parameters, metrics and artifacts enabling reproducible development.
- Designed visualization dashboard with **Matplotlib** and **Seaborn** displaying feature importance and prediction confidence consistently.

PROJECTS

Syllabus-to-Schedule: Intelligent Academic Parser & ETL Pipeline | Python, Gemini, GPT-4o, PostgreSQL, Docker

- Architected LLM-native ETL pipeline using **Gemini 2.0 Flash** extracting structured schedules from PDFs and images with VLM reasoning.
- Engineered dual-stage filtering and conflict resolution engine with **GPT-4o** using Trust Hierarchy achieving 100% logical consistency.
- Implemented **CI/CD** workflows with **GitHub Actions** validating schemas and executing evaluations blocking regressions on golden set.
- Deployed pipeline with **Docker** containers and **PostgreSQL** database processing 1K+ documents per day with sub-2s latency.
- Established monitoring with **Prometheus** tracking extraction accuracy, latency and cost enabling continuous performance optimization.

Artified Self: Multimodal AI Personal Narrative Engine | Python, TensorFlow, PostgreSQL, Redis, FastAPI, Kubernetes

- Designed E2E multimodal ML pipeline ingesting high-frequency telemetry performing temporal segmentation and embedding generation.
- Constructed production inference stack integrating **Google Calendar** APIs, embedding services and LLM generation monitoring latency.
- Implemented data processing pipeline with **Pandas** and **PySpark** transforming 10M+ events per day into structured life-log records.
- Deployed services with **Kubernetes** on **AWS EKS** implementing auto-scaling and health checks ensuring high availability.
- Established **Redis** caching layer for frequent queries reducing database load by 75% and improving API response time significantly.

Sora Persona Engine: AI-Powered Prompt Customization System | Python, SQL, FastAPI, Docker, Airflow

- Engineered data collection pipeline scraping large-scale prompt data via Chrome extension storing structured histories in **PostgreSQL**.
- Constructed prompt-optimization service exposing distilled strategies through **FastAPI** REST APIs with authentication and rate limiting.
- Implemented batch LLM analysis with **Airflow** orchestrating data processing, model inference and quality evaluation workflows.
- Deployed service with **Docker** containers on **AWS EC2** implementing load balancing and monitoring handling 5K+ requests per day.
- Established evaluation framework tracking prompt quality, generation latency and user satisfaction enabling continuous improvement.